Dynamic Hand Gesture Recognition Based on Micro-Doppler Radar Signatures Using Hidden Gauss–Markov Models

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***Abstract*— Dynamic hand gesture recognition using the microwave or millimeter-wave radar sensors has become a typical technology for many human-computer interaction (HCI) applications. In this letter, a novel method is proposed for dynamic hand gesture recognition based on micro-Doppler radar signatures. The short-time Fourier transform is carried out on the raw data to obtain the time-frequency spectrogram. The time-frequency spectrograms associated with the same dynamic hand gesture are modeled by a hidden Gauss–Markov model (HGMM), and the testing gesture is recognized by the maximum likelihood criterion. Experimental results with real radar data demonstrate that the proposed method has a strong generalization ability for radar gesture recognition in the cases of low signal-to-noise ratio (SNR) and unknown users.**

***Index Terms*— Dynamic hand gesture recognition, hidden Markov model (HMM), micro-Doppler, radar sensor.**

I. Introduction

THE hand gesture is deemed to be a natural way of human-computer interaction (HCI). Gesture acquisition devices based on optical or wearable sensors are either susceptible to light conditions or cumbersome for routine use [1], [2]. In contrast, radar sensors can provide rich Doppler information regardless of the illumination condition, and the kinematics associated with different dynamic hand gestures generate distinguishable micro-Doppler signatures [3]–[6]. Thus dynamic hand gesture recognition using the microwave or millimeter-wave radar sensors is becoming a new trend of HCI [7], [8].

The radar sensors frequently used for gesture recognition include a frequency-modulated continuous-wave (FMCW) radar and a continuous-wave (CW) radar [9]. The latter is preferable than the former from the viewpoint of hardware cost. The collection of training data is usually carried out in short distances to guarantee sufficient energy reflection from the hand. However, in the cases of gesture sensing in a relatively long distance, the signal-to-noise ratio (SNR) of testing samples may be much smaller than that of the training samples. Thus it is worth investigating how to improve the generalization ability of radar gesture sensing methods in low SNR cases.

Generally, dynamic hand gesture recognition using a radar sensor consists of two main steps: 1) feature extraction and 2) classification. In step 1), a series of features are extracted from the raw radar data via certain feature extraction techniques. In [10], the principal component analysis (PCA) has been adopted to extract the main features of the micro-Doppler signatures. By exploiting the intrinsic sparsity of gesture signal collected by a CW radar, a sparsity-driven method was devised to extract the time-frequency trajectories of dynamic hand gestures [11]. Other feature extraction techniques can be found in [12] and references therein. In step 2), the extracted features are fed into a pretrained classifier to decide the class label of the gesture. Typical classifiers consist of k-nearest neighbor (kNN), support vector machine (SVM), backpropagation neural network (BPNN), and so on [13].

Hidden Markov model (HMM) [14] is a powerful probabilistic approach for modeling time-varying sequences. An HMM is viewed as a doubly embedded stochastic process, i.e., an invisible Markov chain which generates the sequence of hidden states, and a visible process which generates the sequence of observations. Moreover, the probability distribution of an observation depends only on the associated state. In essence, the movements of fingers and palm can be treated as “hidden states” in a Markov model, which allows us to formulate the micro-Doppler signatures of dynamic hand gestures under the HMM framework.

In this letter, we intend to adopt HMMs for dynamic hand gesture recognition using only micro-Doppler signatures. Our method is based on the hidden Gauss–Markov models (HGMMs) [15], in which the state vectors, as well as the observation vectors, take values from continuums. In the proposed method, the raw radar data of a dynamic hand gesture is first processed by a short-time Fourier transform (STFT) to generate a sequence of frequency spectrogram vectors (FSVs). Then the FSV sequences from the same gesture are used to train the associated HGMM. In the testing phase, the likelihoods of the testing signal under all the trained HGMMs are compared, and the recognition is accomplished by finding the class label corresponding to the largest likelihood. Experimental results based on real data demonstrate that the proposed method has a strong generalization ability in the case of low SNR and unknown users.

Manuscript received July 23, 2019; revised December 12, 2019; accepted February 15, 2020. Date of publication February 26, 2020; date of current version January 21, 2021. This work was supported by a grant from the Institute for Guo Qiang, Tsinghua University. *(Corresponding author: Gang Li.)*

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Digital Object Identifier 10.1109/LGRS.2020.2974821

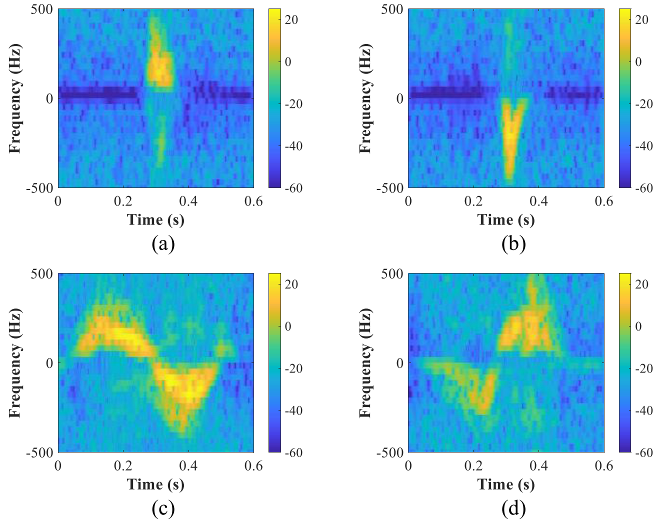

Fig. 1. Spectrograms of four dynamic hand gestures performed by one subject. (a) Holding. (b) Pushing. (c) Rotating clockwise. (d) Rotating counterclockwise.

The rest of this letter is organized as follows. The radar data collection is summarized in Section II. In Section III, the proposed method is formulated in detail. Experiments on the measured data are presented in Section IV. Finally, Section V concludes the letter.

II. Data Collection

The hand gesture data were collected by a CW K-band radar (Ancortek SDR-kit 2500B), as described in [16]. The radar consists of a pair of transmitting and receiving horn antennas both with a beamwidth of 60° in the horizontal and vertical directions. The radar operates at 25 GHz with a baseband sampling frequency of 1 kHz and output power of 16 dBm. Four dynamic hand gestures, i.e., 1) holding, 2) pushing, 3) rotating clockwise, and 4) rotating counterclockwise, were considered. The gestures were performed in front of the radar antennas at a distance of about 0.3 m. Five subjects participated in the data collection. For each gesture, 20 recordings per subject were collected, thus yielding a total number of 400 recordings. For each recording, the observation time was 0.6 s.

The time-frequency spectrograms of four dynamic hand gestures performed by one subject are shown in Fig. 1, where the stationary clutter has been suppressed before performing STFT. Obviously, the time-frequency spectrograms of the four dynamic hand gestures are significantly different, which implies the potential possibility of distinguishing them.

III. Proposed Method

Fig. 2 illustrates the processing scheme of the proposed method. At first, the raw radar data of the training samples are preprocessed to generate FSV sequences. Then the HGMMs are trained by these FSV sequences. In the testing stage, the FSV sequence of a testing sample is fed into the HGMM classifier to decide its class label. Detailed descriptions of the proposed method are given below.

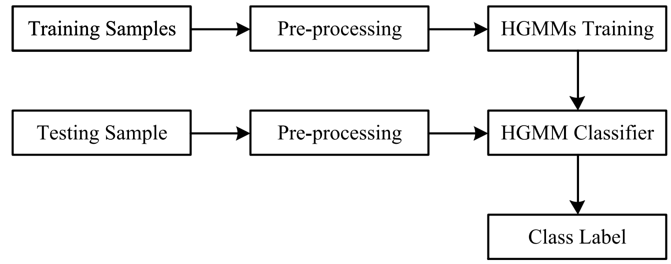

Fig. 2. Processing scheme of the proposed method.

A. Preprocessing

The recorded data contain strong echoes reflected from a static human torso, which may interfere with the echoes reflected from the gesturing hand, and thus, should be eliminated before recognition. Let $s(t)$, $t = 1, \ldots, T$, denote the recorded raw radar data, where T is the total number of sampling points. The stationary component can be canceled by the classical moving target indication (MTI) approach, that is

$$g(t) = s(t) + s(t+2) - 2s(t+1), \quad t = 1, \ldots, T-2. \tag{1}$$

Then the STFT is applied to $g(t)$ to obtain the micro-Doppler signatures, that is

$$G(m,n) = \sum_{t=1}^{M} w(t)\, g[t + (n-1)h]\, e^{-\mathrm{j}\frac{2\pi t}{M}\left(m - \frac{M}{2}\right)} \tag{2}$$

where $w(t)$ is a window function, M represents the window length, h stands for the sliding step width, $m = 1, \ldots, M$, $n = 1, \ldots, N$, with $N = \lfloor (T - M - 2)/h \rfloor + 1$, and $\lfloor \cdot \rfloor$ denotes the operator of taking the lower integer value. In this letter, we fix $M = 32$, $h = 16$, $N = 36$, and use a hamming window.

After taking the logarithm of the absolute value of $G(m,n)$, that is

$$Z(m,n) = 10 \log(|G(m,n)| + \delta) \tag{3}$$

where δ is a small positive scalar to keep the value from being too small, and it is set to be 10^{-6} in this letter, we define $\boldsymbol{z}_n = [Z(1,n), \ldots, Z(M,n)]^T$ as an FSV and define $\boldsymbol{Z} = \{\boldsymbol{z}_1, \ldots, \boldsymbol{z}_N\}$ as an FSV sequence, where $(\cdot)^T$ represents the transpose operator. We also normalize the elements in $\boldsymbol{Z}$ to be within [0, 1] to eliminate the varying strengths of different recordings.

B. Training Phase

The concept of HGMM is extended from the basic HMM by allowing the hidden states to assume values on a continuum, as well as supposing the probability functions are of Gaussian form [15]. A graphical representation of HGMM is given in Fig. 3, where the squares and circles represent hidden states and observations, respectively.

To fit the problem, in our case, we denote the hidden state sequence as $\boldsymbol{X} = \{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_N\}$, where $\boldsymbol{x}_n \in \mathbb{R}^L$ and L is the dimension of the state vector. As indicated in Fig. 3, the state vectors follow a first-order Markov process, such

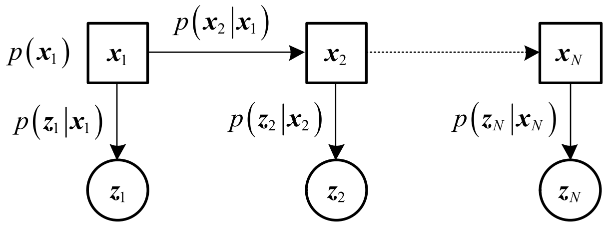

Fig. 3. Graphical representation of HGMM.

that $p(\boldsymbol{x}_n|\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_{n-1}) = p(\boldsymbol{x}_n|\boldsymbol{x}_{n-1})$ for $n = 2, \ldots, N$, with $p(\boldsymbol{x}_1)$ being the initial state probability density. The vectors in the observation sequence, i.e., $\boldsymbol{Z} = \{z_1, \ldots, z_N\}$, in our case, are conditionally independent, given the states.

An HGMM is thus fully characterized by a set of distributions, i.e., the *a priori* distribution for the initial state, the state-transition distributions, and the observation output distributions. Furthermore, these distributions are specialized to the following linear Gaussian densities, respectively:

$$p(\boldsymbol{x}_1; \theta_1) = \mathcal{N}(\boldsymbol{x}_1; \boldsymbol{\mu}_1, \boldsymbol{P}_1) \tag{4}$$

$$p(\boldsymbol{x}_n|\boldsymbol{x}_{n-1}; \theta_X) = \mathcal{N}\left(\boldsymbol{x}_n; \boldsymbol{A}_n\boldsymbol{x}_{n-1}, \boldsymbol{Q}_n\right) \tag{5}$$

$$p(z_n|\boldsymbol{x}_n; \theta_Z) = \mathcal{N}(z_n; \boldsymbol{B}_n\boldsymbol{x}_n, \boldsymbol{R}_n) \tag{6}$$

where $\mathcal{N}(\boldsymbol{y}; \boldsymbol{\mu}, \boldsymbol{P})$ represents the Gaussian density for vector $\boldsymbol{y}$ with the mean vector $\boldsymbol{\mu}$ and the nonsingular covariance matrix $\boldsymbol{P}$, $\theta_1 = \{\boldsymbol{\mu}_1, \boldsymbol{P}_1\}$, $\theta_X = \{(\boldsymbol{A}_n, \boldsymbol{Q}_n), n = 2, \ldots, N\}$, and $\theta_Z = \{(\boldsymbol{B}_n, \boldsymbol{R}_n), n = 1, \ldots, N\}$. Obviously, an HGMM is parameterized by $\{\theta_1, \theta_X, \theta_Z\}$. The expectation-maximization (EM) algorithm has been formulated in [15] to estimate the HGMM parameters given the observation sequences.

In the proposed method, each class of dynamic hand gestures is modeled by an HGMM. Specifically, the FSV sequence associated with a dynamic hand gesture is recognized as an observation sequence generated by an HGMM, and the FSV sequences obtained from the training samples related to this particular class of dynamic hand gestures are utilized to estimate the HGMM parameter set. For convenience, relative to a C-class problem, we denote $\lambda^c = (\theta_1^c, \theta_X^c, \theta_Z^c)$ as the HGMM parameter set associated with category c, where $c = 1, \ldots, C$.

As shown in Fig. 2, after preprocessing the training samples, the EM algorithm is employed to estimate the parameter set of each HGMM. Then the trained HGMMs are utilized to build up the HGMM classifier.

C. Testing Phase

For a testing sample to be classified, after the same preprocessing as the training stage, its FSV sequence is inputted into the HGMM classifier to determine its class label by identifying the HGMM that can best describe the FSV sequence. In the HGMM framework, this problem can be solved by scoring the probability that the observation sequence $\boldsymbol{Z}$ is generated by the HGMM λ^c, that is

$$p\left(\boldsymbol{Z}|\lambda^c\right) = \int d\boldsymbol{X}\, p\left(\boldsymbol{Z}, \boldsymbol{X}; \lambda^c\right). \tag{7}$$

In (7), the symbol $\int d\boldsymbol{X}$ stands for the multiple integral $\int d\boldsymbol{x}_1 \ldots \int d\boldsymbol{x}_N$, where each single integral $\int d\boldsymbol{x}_n$ is a multidimensional integration over the state space; $p(\boldsymbol{Z}, \boldsymbol{X}; \lambda^c)$ denotes the joint probability of $\boldsymbol{Z}$ and $\boldsymbol{X}$, and it is cast as

$$p\left(\boldsymbol{Z}, \boldsymbol{X}; \lambda^c\right) = p\left(\boldsymbol{x}_1\right) p\left(z_1|\boldsymbol{x}_1\right) \prod_{n=2}^{N} p\left(\boldsymbol{x}_n|\boldsymbol{x}_{n-1}\right) p\left(z_n|\boldsymbol{x}_n\right) \tag{8}$$

where the explicit parameters, i.e., θ_1^c, θ_X^c, and θ_Z^c, on the right-hand side of (8) are dropped for convenience. Equation (7) can be computed by the forward-backward algorithm formulated in [15]. Finally, the maximum likelihood criterion is employed for the class label assignment task, that is

$$\text{Class}\,(Z) = \arg\max_{c}\left(p\left(Z|\lambda^c\right)\right). \tag{9}$$

IV. Experimental Results

In this section, we assess the performance of the proposed HGMM-based method in terms of recognition accuracy. The PCA-based methods, the sparsity-driven method [11], as well as the long short-term memory (LSTM) method [17] are used for comparison. For the proposed method, the iteration number of the EM algorithm is set as 30, the HGMM parameters are initialized as follows: $\boldsymbol{\mu}_1$ is initialized as a full-zero vector; $\boldsymbol{A}_n$, $n = 2, \ldots, N$, and $\boldsymbol{B}_n$, $n = 1, \ldots, N$, are initialized as random matrices; $\boldsymbol{P}_1$, $\boldsymbol{Q}_n$, $n = 2, \ldots, N$, and $\boldsymbol{R}_n$, $n = 1, \ldots, N$, are initialized as scaled identity matrices with scale factor 10^5. For the PCA-based methods, we first vectorize each training FSV sequence and then stack them into a data matrix; the largest five principal components of the data matrix form the feature extraction matrix; the kNN and BPNN classifier are adopted for recognition. For the PCA-kNN, the number of neighbors is set as five, and the Euclidean distance is adopted. For the PCA-BPNN, the hidden layer has ten neurons. For the sparsity-driven method, the sparsity level is set to be 17, while other parameters are the same as that used in [11]. For the LSTM method, the number of hidden units is 100, and the FSV sequences are taken as input.

A. Effect of State Vector Dimension on Recognition Accuracy

In the first experiment, we investigate the effect of the state vector dimension L on the recognition performance of the proposed method. The state vector dimension takes value from 4 to 16, with an increased step of 2. For each gesture, we randomly select 40 recordings from the related data as training samples, with the rest of the data as testing samples. To be precise, the training samples for each gesture are equally selected from the subjects, and the training samples from each subject are randomly selected. The experimental results averaged over 100 Monte Carlo runs are illustrated in Fig. 4.

As revealed in Fig. 4, the HGMM-based method achieves very stable recognition accuracies for different state vector dimensions. This indicates that the HGMM-based method is not sensitive to the value of the state vector dimension. In the following experiments, we fix the state vector dimension as four.

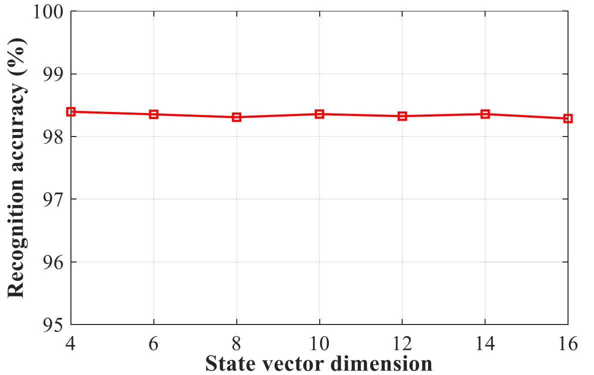

Fig. 4. Recognition accuracy versus state vector dimension for the HGMM-based method in the case of 40 training samples for each gesture.

B. Effect of Noise Level on Recognition Accuracy

As introduced in Section II, the raw data were collected by performing dynamic hand gestures just in front of the radar. On the other hand, the need for long-distance applications exist in practice. Longer distance implies lower SNR; thus, we investigate the recognition accuracies of the methods under different noise levels. To simulate low SNR data, we treat the raw data collected in our experiments as pure signals and add different levels of white Gauss noise to them in the following way:

$$\boldsymbol{s'}_{i,j,c} = \boldsymbol{s}_{i,j,c} + \boldsymbol{n} \tag{10}$$

where $\boldsymbol{s}_{i,j,c}$ stands for the ith vector-form recorded raw data that belongs to class c and performed by subject j, $\boldsymbol{n}$ represents the added white Gauss noise, $\boldsymbol{s'}_{i,j,c}$ denotes the simulated low SNR gesture data. For ease of exposition, we define the SNR as

$$\text{SNR} = \frac{\sum_{c=1}^{C}\sum_{j=1}^{J}\sum_{i=1}^{I}\|s_{i,j,c}\|_2^2}{CJI\|n\|_2^2} \tag{11}$$

where I, J, and C represent the number of recordings for each subject and each class, the number of subjects, and the number of classes, respectively, and $\|\cdot\|_2$ denotes the L_2 norm.

In this experiment, we randomly select 60 samples for each gesture to generate the training set, while the remaining samples are used to generate the testing set, and the noise is only added to the testing samples. The recognition results based on 100 Monte Carlo trials are plotted in Fig. 5.

From Fig. 5 it is clear that the recognition accuracies for each method are improved with increased SNR. Notably, the HGMM-based method outperforms all other methods in all the SNR settings. The reason is that the probability function in HGMMs encompasses noise modeling, which makes the HGMM-based method more robust to noise. However, for the other methods, the features extracted from a low SNR testing sample can deviate a lot from the features extracted from the training samples of the same class, so degraded recognition performances are induced. The results in Fig. 5 indicate that the HGMM-based method has much better generalization ability in low SNR situations, which makes it more suitable for the applications of long-distance gesture sensing.

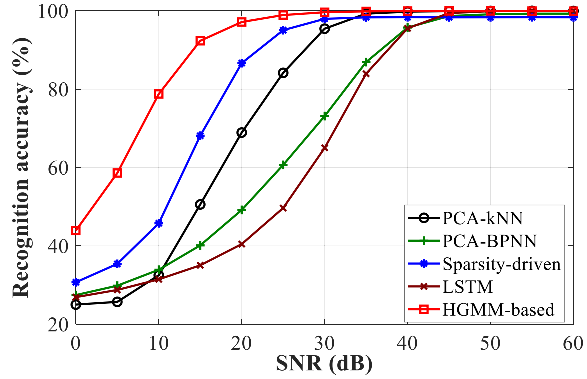

Fig. 5. Recognition accuracies under different noise levels.

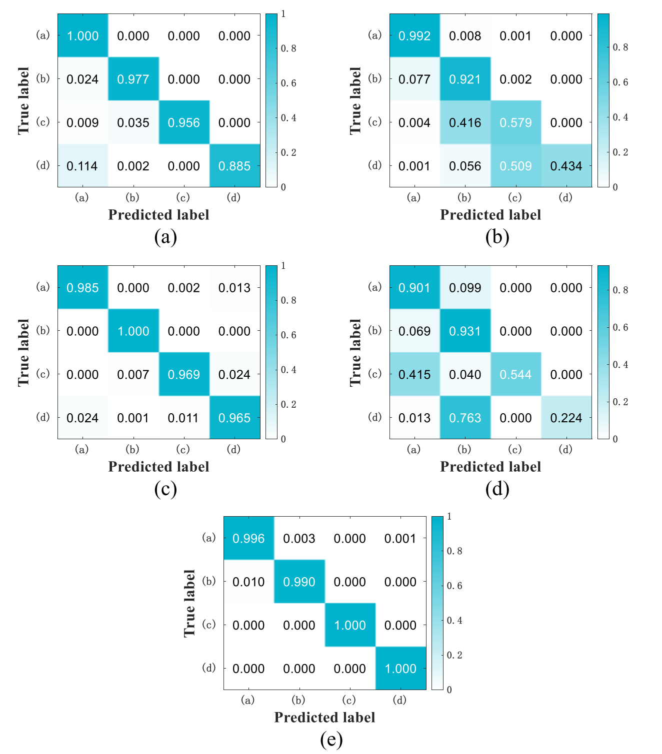

Fig. 6. Confusion matrices under the case of 30-dB SNR for each gesture. (a) PCA-kNN. (b) PCA-BPNN. (c) Sparsity-driven. (d) LSTM. (e) HGMM-based.

TABLE I
TIME CONSUMPTION FOR EACH METHOD

Method	PCA-kNN	PCA-BPNN	Sparsity-Driven	LSTM	HGMM-based
Time	1.8 ms	9.6 ms	26.2 ms	6.2 ms	21.2 ms

To more clearly visualize the recognition results, the confusion matrices of these methods under the case of 30-dB SNR are drawn in Fig. 6.

As depicted in Fig. 6, the confusion matrix of the HGMM-based method exhibits a strong diagonal pattern, indicating that all of these four gestures can be accurately classified. It can also be seen that gestures (c) and (d) tend to be misclassified by other methods in the case of low SNR. The reason lies in the fact that the features extracted by other methods for gestures (c) and (d) are more likely to be

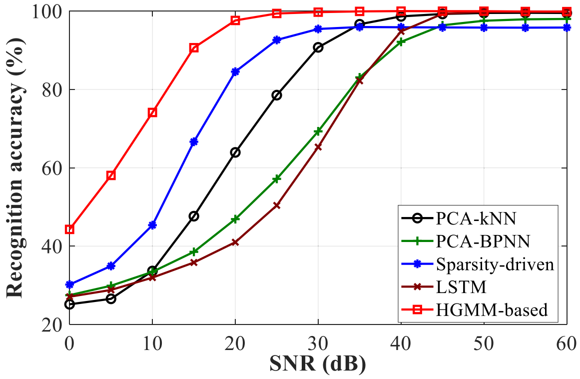

Fig. 7. Recognition accuracies for unknown users under different noise levels.

contaminated by noise as these two gestures last for longer time intervals.

C. Recognition Accuracy for Unknown Users

In this experiment, we aim to verify the performance of the HGMM-based method in recognizing dynamic hand gestures of unknown users. For doing so, the recordings from three subjects are used to construct the training set, and the recordings from the other two subjects are used to construct the testing set. Therefore, totally, there are ten combinations for data set partition. Only the testing samples are added with noise. The numerical results obtained by averaging over 100 Monte Carlo trails (ten for each combination), are given in Fig. 7.

As exposed in Fig. 7, the HGMM-based method gets the best recognition accuracy among all these methods under all the SNR settings. By comparing the curves in Fig. 7 and that in Fig. 5, it is easy to see that the HGMM-based method does not show significant performance differences in the cases of known and unknown users.

D. Analysis About the Time Complexity

The computational time elapsed by each method for testing one sample is reported in Table I, where the results are averaged over 100 trials. The hardware platform is a desktop PC equipped with an Intel Core i5-8500 CPU and a 16-GB RAM running MATLAB 2019a on Windows 10.

It can be seen that the HGMM-based method consumes more time than the PCA-kNN, PCA-BPNN, and LSTM methods, but it consumes less time than the sparsity-driven method.

V. Conclusion

An HGMM-based method is proposed for recognizing dynamic hand gestures using micro-Doppler radar signatures. Experiments on real radar data have demonstrated that the proposed method has a strong generalization ability in the cases of low SNR and unknown users. In the future, we would like to investigate the problem of radar-based dynamic hand gesture recognition in more complicated situations by appealing to the outstanding modeling capability of HGMM.

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